Result 1 (identity components commute). [Topology foundations, diagram map laws]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let X be a module diagram on Q. Then

$$\mathrm{id}(X) \text{ commutes with every edge from } X \text{ to } X.$$

Result 2 (composite components commute). [Topology foundations, diagram map laws]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let X, Y and Z be module diagrams on Q. Let f be a map of module diagrams from X to Y. Let g be a map of module diagrams from Y to Z. Then

$$(g \circ f) \text{ commutes with every edge from } X \text{ to } Z.$$

Result 3 (identity diagram component). [Topology foundations, diagram map laws]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let X be a module diagram on Q. Let h be evidence that $\mathrm{id}(X)$ commutes with every edge from X to X. Let v be an element of V. Then

$$\mathrm{id}(X).\mathrm{component}(v) = \mathrm{id}_{X.\mathrm{object}(v)}\,.$$

Result 4 (composite diagram component). [Topology foundations, diagram map laws]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let X, Y and Z be module diagrams on Q. Let f be a map of module diagrams from X to Y. Let g be a map of module diagrams from Y to Z. Let h be evidence that $(g \circ f)$ commutes with every edge from X to Z. Let v be an element of V. Then

$$(g \circ f).\mathrm{component}(v) = g.\mathrm{component}(v) \circ f.\mathrm{component}(v).$$

Result 5 (diagram map extensionality). [Topology foundations, diagram map laws]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let X and Y be module diagrams on Q. Let f and g be elements of $\mathrm{Hom}(X, Y)$. Assume (for every $v : V,\ f.\mathrm{component}(v) = g.\mathrm{component}(v)$) (`components`). Then

$$f = g.$$

Result 6 (identity diagram exists). [Topology foundations, diagram map laws]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let X be a module diagram on Q. Then

$$\big(\text{there exists } f : \mathrm{Hom}(X, X),\ \big(\text{for every } v : V,\ f.\mathrm{component}(v) = \mathrm{id}_{X.\mathrm{object}(v)}\big)\big)\,.$$

LLM-assisted specimens and proofs (gpt-6, 2026-11-06). See PROVENANCE.md. Diagram maps; the pyramid construction remains pending.